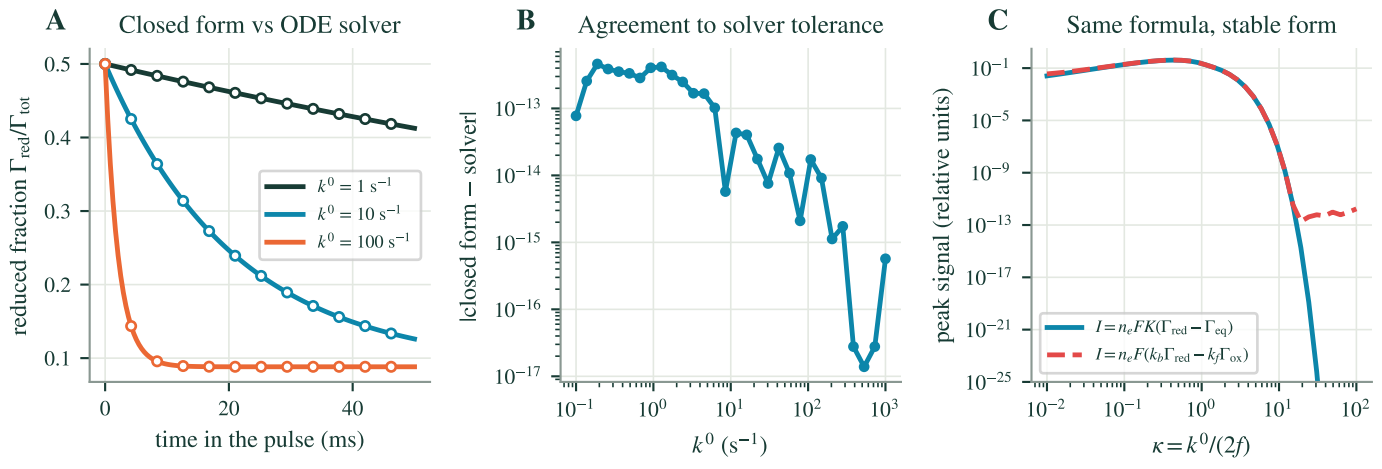
A
Closed form vs ODE solver
reduced fraction $\Gamma_{red}/\Gamma_{tot}$
0.5
0.4
0.3
0.2
0.1
0
20
40
time in the pulse (ms)
$k^0 = 1\ s^{-1}$
$k^0 = 10\ s^{-1}$
$k^0 = 100\ s^{-1}$
B
Agreement to solver tolerance
|closed form − solver|
10^{-13}
10^{-14}
10^{-15}
10^{-16}
10^{-17}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
k^0 (s^{-1})
C
Same formula, stable form
peak signal (relative units)
10^{-1}
10^{-5}
10^{-9}
10^{-13}
10^{-17}
10^{-21}
10^{-25}
$I = n_e F K(\Gamma_{red} - \Gamma_{eq})$
$I = n_e F(k_b \Gamma_{red} - k_f \Gamma_{ox})$
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
$\kappa = k^0/(2f)$